

HCI / UX STUDIO

Week 7–8 Usability Testing Worksheet

Peer Evaluation · Discussion · Feedback Consolidation

Your Name: Nhean Omra

Your Team: 12

Team Reviewed: 13

HOW TO USE THIS WORKSHEET

PART 1 Individual Evaluation	Work independently. Observe the prototype and answer all questions in writing. Be specific — cite what you saw or experienced.
PART 2 Group Discussion	Share your written answers with the other team. Use the discussion prompts to guide a structured conversation. Record what was said.
PART 3 Consolidation	Back as your own team. Synthesize the feedback you received, prioritize what to act on, and plan your next design iteration.

PART 1 — INDIVIDUAL USABILITY EVALUATION

Complete this section on your own before discussing it with the other team. Be analytical and specific in every answer.

A. First Impressions

☐ *Think-aloud: narrate silently as you explore. What is your immediate mental model of what this app does?*

1. In one sentence, what problem does this prototype appear to solve — and for whom?

The problem that the prototype helps solve is for students to visualise real world experiments with theory in class. This is for high school students.

2. Without any explanation, what is the first action you would take? What visual or structural cue led you there?

B. Usability Heuristics Assessment

☐ *Based on Nielsen (1994). For each heuristic, rate how well the prototype satisfies it, then justify with a specific example.*

Heuristic	1	2	3	4	5
Visibility of system status	●	●	●	●	●
Match between system & real world	●	●	●	●	●
User control & freedom	●	●	●	●	●
Consistency & standards	●	●	●	●	●
Error prevention	●	●	●	●	●
Recognition over recall	●	●	●	●	●
Flexibility & efficiency of use	●	●	●	●	●
Aesthetic & minimalist design	●	●	●	●	●

Scale: 1 = Major issue 2 = Problematic 3 = Minor issue 4 = Adequate 5 = Well handled

3. Choose the TWO lowest-rated heuristics above. For each, describe a specific moment in the prototype where this heuristic was violated and explain why it could confuse users.

Heuristic 1: Flexibility & efficiency of use: student might get confused or might not find what they wanted, as there is not much description explaining what

Heuristic 2: match between real world and system: some learning formula might be irrelevant with the school curriculum and learning program

C. Task Flow & Information Architecture

☐ *Think about the journeys a user would take through the app — not just individual screens.*

4. Describe the primary user task this prototype supports. Map the steps you took to complete it (or as far as you got). Where did you hesitate or have to backtrack?

User task in this prototype is well design. No unnecessary log in, no subscription, just launch simulation and test the formula.

5. Was there a moment where you felt lost or unsure of where to go next? Describe the screen or transition and what information was missing to guide you.

System properly display direction with smooth and responsive transition

6. Did the prototype hide important features or overload any single screen with too many options? Cite a specific example and suggest a structural alternative.

Features and formulas are well displayed for easy use so it is not as confusing to browse through

D. Real-World Problem Fit

☐ *Evaluate how well the design addresses the actual problem it claims to solve.*

7. Who is the intended user of this app? Does the design language, terminology, and feature set match what that user would realistically know and need? Explain.

Compare user to this app intend, i think it matches well, although some formulas might not be relevant to the learning program of the intended user

8. What real-world constraint (time pressure, distraction, limited connectivity, emotional state) is NOT accounted for in this prototype? How might that undermine usability?

From the demonstration so far there is not much constraint apart from accessibility as it requires internet to access the website

9. Does the prototype solve the root problem, or does it address a symptom? Justify your answer with reference to specific design choices you observed.

The prototype does both solve the problem and address a symptom as the problem is not as in need of solving but can help students understand more of what they are learning

E. Prototype Quality & Fidelity

10. What is the fidelity level of this prototype (low / mid / high)? Is the fidelity appropriate for what they are trying to learn at this stage? Why or why not?

The fidelity level is high as it is as close to the final project as they only need to finish the formula simulation

11. What is the single most impressive design decision in this prototype? What evidence suggests it was intentional rather than accidental?

The most impressive design is the functionality and the options between 3 big subject in highschool

12. If you had to prioritize ONE change that would most improve this prototype before the next iteration, what would it be and why?

It would be an option where student can paste their own equation into the formula for instant simulation rather than having to scroll through the wheel for their equations

PART 2 — INTER-TEAM DISCUSSION

Each team shares their written evaluations. Use these structured prompts to guide your discussion. Assign a note-taker. Record key points — agreements, surprises, and disagreements.

Before You Start

Presenting Team

Listen actively. Do not defend or explain your design unless asked a clarifying question. Resist the urge to justify — just listen and take notes.

Reviewing Team

Deliver feedback as an observer, not a competitor. Use specific evidence. Separate what you observed from your interpretation.

Discussion Prompts — Work through these in order

D1. What was the core user problem this prototype is trying to solve? Did both teams interpret it the same way?

☐ *Surface any mismatch between intent (presenting team) and perception (reviewing team). Note: this is about communication, not correctness.*

Notes from discussion:

Both teams agreed that the prototype aims to help high school students better understand scientific concepts through interactive simulations.

- The reviewing team initially thought the system was only a formula calculator, while the presenting team explained that the main goal is interactive learning and visualization.
- Overall, both teams shared a similar understanding after the explanation

D2. Which usability heuristic violations had the biggest overlap between your two written evaluations? Which had the biggest disagreement?

☐ *Use your heuristic rating tables. Discuss why two observers rated the same thing differently — this is often the most valuable insight.*

Notes from discussion:

Both teams agreed that flexibility and efficiency could be improved by allowing users to search or directly input formulas.

- Both teams also suggested providing more guidance for first-time users.
- The biggest disagreement was on the match between the real world and the system. The reviewing team believed the formulas were sufficient, while the presenting team explained they were based on the school curriculum.

D3. Reviewing team: describe one moment where a user might abandon the app entirely. Presenting team: was that scenario considered during design?

☐ *Focus on friction points that cross a threshold — not just annoyance but task failure.*

Notes from discussion:A new user may leave the application if they cannot quickly locate the formula or simulation they need.

- The presenting team acknowledged this issue and stated they plan to improve navigation by adding better categorization and search functionality.

D4. Does the design reflect a real understanding of the target user, or does it reflect what the design team assumed the user would want?

☐ *Look for evidence of user research vs. assumption. Have specific screens or features to cite.*

Notes from discussion:

The design generally reflects the needs of high school students.

- However, some assumptions were made regarding which formulas students use most frequently.
 - More user feedback from actual students would help validate these assumptions.
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D5. What is one design decision in this prototype that surprised you in a positive way — something you would not have thought of yourself?

☐ *Genuine appreciation develops shared design vocabulary and helps teams identify their own strengths.*

Notes from discussion:

The interactive simulations across different science subjects were impressive.

- The clean interface and responsive animations created an engaging learning experience.

D6. If this prototype were released tomorrow, what is the most likely failure mode — technical, usability, or problem-fit?

☐ *This is a synthesis question. Push for specificity: what would actually go wrong for a real user?*

Notes from discussion:

- The biggest risk would be usability rather than technical issues.
- Students may struggle to locate specific formulas if the content library continues to grow without improved organization.

- Limited offline accessibility could also reduce usability in schools with unstable internet connections.

PART 3 — FEEDBACK CONSOLIDATION & ITERATION PLAN

Return to your own team. Review everything written and discussed today. Work together to complete this section, it becomes your roadmap for the next design sprint.

Step 1 — Inventory the Feedback

List all pieces of feedback received (from written evaluations and discussion). Do not filter yet — capture everything.

Strengths identified by reviewers	Issues / concerns raised
1.Clean and modern interface. 2.Interactive simulations improve learning. 3.Easy-to-understand navigation. 4.Smooth page transitions. 5.Well-organized scientific topics. 6.High-fidelity prototype with professional appearance.	1.No search function for formulas. 2.Limited explanation for some formulas. 3.Some formulas may not match every school curriculum. 4.Internet connection is required. 5.Navigation may become difficult as more content is added. 6.Users cannot input their own formulas.

Step 2 — Prioritization Matrix

For each issue from Step 1, rate its Impact (how much it affects user success) and Effort to fix (time/complexity). Then plot it in the grid below.

	Low Effort	High Effort	Issue #s placed here
High Impact	✔ Do first Issues:	❑ Plan carefully Issues:	Issue #1: Add formula search feature. • Issue #2: Add brief explanations for each formula.

Low Impact	☐ Schedule Issues:	☒ Deprioritise Issues:	Issue #3: Review curriculum relevance. • Issue #5: Improve content categorization.
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Step 3 — Analytical Reflection

R1. What feedback surprised your team the most? Why did that gap exist between your intention and the reviewer's experience?

The suggestion to allow users to enter their own formulas was the most surprising. We originally focused on providing predefined simulations, but reviewers pointed out that custom input would make the system more flexible and useful for advanced learners.

R2. Which piece of feedback do you most disagree with, and why? What would you need to see to change your mind?

We slightly disagree with the suggestion that the selected formulas are not relevant. Our formula selection was based on commonly taught high school science topics. However, we recognize that different schools may have different curricula, so additional research would help validate our selection.

R3. Looking at your prototype through Nielsen's heuristics, which one principle should anchor your next redesign? Explain the connection to specific feedback you received.

Flexibility and Efficiency of Use should guide the next redesign. Feedback consistently suggested adding search functionality, better navigation, and allowing users to access formulas more quickly, which would significantly improve efficiency.

R4. If you were to conduct a formal think-aloud usability test next week (NN/g protocol), what specific task scenario would you give a user? Write the task prompt below.

"You have a physics assignment about projectile motion. Use the application to locate the correct simulation, explore the formula, and complete a simulation using the provided values. While completing the task, explain what you are thinking and describe any confusion or difficulty you experience."

Step 4 — Iteration Action Plan

Your team commits to the following changes before the next session:

#	Change / Action	Linked Feedback	Owner & By When
1	Add formula search functionalit	Issue #1	
2	Add short explanations for every formula	Issue #2	
3			
4			
5			

Step 5 — Team Sign-off

Each team member signs to commit to the iteration plan.

Name: ____Nhean Omra_____	Name: ____Ngoun Sovichet _____	Name: ____Lim vinchay_____
Signature: _____	Signature: _____	Signature: _____

References: Nielsen, J. (1994). Usability inspection methods. | Buchenau & Suri (2000). Experience prototyping. | NN/g Usability Testing 101.